CLASS X : CHAPTER - 15
PROBABILITY

IMPORTANT FORMULAS & CONCEPTS

PROBABILITY

Experimental or empirical probability P(E) of an event E is

$$P(E) = \frac{\text{Number of trials in which the event happened}}{\text{Total number of trials}}$$

The theoretical probability (also called classical probability) of an event A, written as P(A), is defined as

$$P(A) = \frac{\text{Number of outcomes favourable to A}}{\text{Number of all possible outcomes of the experiment}}$$

Two or more events of an experiment, where occurrence of an event prevents occurrences of all other events, are called **Mutually Exclusive Events.**

COMPLIMENTARY EVENTS AND PROBABILITY

We denote the event 'not E' by $\bar{E}$. This is called the **complement** event of event E.

So, P(E) + P(not E) = 1

i.e., P(E) + P($\bar{E}$) = 1, which gives us P($\bar{E}$) = 1 – P(E).

In general, it is true that for an event E, P($\bar{E}$) = 1 – P(E)

- ☞ The probability of an event which is impossible to occur is 0. Such an event is called an **impossible event**.
- ☞ The probability of an event which is sure (or certain) to occur is 1. Such an event is called a **sure event** or a **certain event**.
- ☞ The probability of an event E is a number P(E) such that $0 \leq P(E) \leq 1$
- ☞ An event having only one outcome is called an elementary event. The sum of the probabilities of all the elementary events of an experiment is 1.

DECK OF CARDS AND PROBABILITY

A deck of playing cards consists of 52 cards which are divided into 4 suits of 13 cards each. They are black spades (♠) red hearts (♥), red diamonds (♦) and black clubs (♣).

The cards in each suit are Ace, King, Queen, Jack, 10, 9, 8, 7, 6, 5, 4, 3 and 2. Kings, Queens and Jacks are called face cards.

Example set of 52 poker playing cards

Suit	Ace	2	3	4	5	6	7	8	9	10	Jack	Queen	King
Clubs													
Diamonds													
Hearts													
Spades													

Equally likely events : Two or more events are said to be equally likely if each one of them has an equal chance of occurrence.

Mutually Exclusive events : Two or more events are mutually exclusive if the occurrence of each event prevents the every other event.

Complementary events : Consider an event has few outcomes. Event of all other outcomes in the sample survey which are not in the favourable event is called Complementary event.

Exhaustive events : All the events are exhaustive events if their union is the sample space.

Sure events : The sample space of a random experiment is called sure or certain event as any one of its elements will surely occur in any trail of the experiment.

Impossible event : An event which will occur on any account is called an impossible event.

MCQ WORKSHEET-I
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PROBABILITY

1. There are 6 marbles in a box with number 1 to6 marked on each of them . What is the probability of drawing a marble with number 2 ?

(a) $\frac{1}{6}$ (b) $\frac{1}{5}$ (c) $\frac{1}{3}$ (d) 1

2. A coin is flipped to decide which team starts the game . What is the probability of your team will start ?

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

3. A die is thrown once . What will be the probability of getting a prime number ?

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

Cards are marked with numbers 1 to 25 are placed in the box and mixed thoroughly. One card is drawn at random from the box. Answer the following questions (Q4-Q13)

4. What is the probability of getting a number 5?

(a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

5. What is the probability of getting a number less than 11?

(a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

6. What is the probability of getting a number greater than 25?

(a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

7. What is the probability of getting a multiple of 5?

(a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

8. What is the probability of getting an even number?

(a) 1 (b) 0 (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

9. What is the probability of getting an odd number?

(a) 1 (b) 0 (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

10. What is the probability of getting a prime number?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

11. What is the probability of getting a number divisible by 3?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

12. What is the probability of getting a number divisible by 4?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

13. What is the probability of getting a number divisible by 7?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

14. A bag has 4 red balls and 2 yellow balls. A ball is drawn from the bag without looking into the bag. What is probability of getting a red ball?

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) 1

15. A bag has 4 red balls and 2 yellow balls. A ball is drawn from the bag without looking into the bag. What is probability of getting a yellow ball?

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) 1

MCQ WORKSHEET-II
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PROBABILITY

A box contains 3 blue, 2 white, and 5 red marbles. If a marble is drawn at *random* from the box, then answer the questions from 1 to 5.

1. What is the probability that the marble will be white?

(a) $\frac{1}{6}$ (b) $\frac{1}{5}$ (c) $\frac{1}{3}$ (d) 1

2. What is the probability that the marble will be red?

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

3. What is the probability that the marble will be blue?

(a) $\frac{3}{10}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

4. What is the probability that the marble will be any one colour?

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

5. What is the probability that the marble will be red or blue?

(a) 1 (b) $\frac{4}{5}$ (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

A die is thrown once, then answer the questions from 6 to 10.

6. Find the probability of getting a prime number

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

7. Find the probability of getting a number lying between 2 and 6

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

8. Find the probability of getting an odd number.

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

9. Find the probability of getting an even number.

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

10. Find the probability of getting a number greater than 4.

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) 1

MCQ WORKSHEET-III
CLASS X: CHAPTER - 15
PROBABILITY

A box contains 5 red marbles, 6 white marbles and 4 green marbles. If a marble is drawn at *random* from the box, then answer the questions from 1 to 6.

1. What is the probability that the marble will be white?

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) 1

2. What is the probability that the marble will be red?

(a) $\frac{1}{6}$ (b) $\frac{2}{3}$ (c) $\frac{1}{3}$ (d) 1

3. What is the probability that the marble will be green?

(a) 0.3 (b) $\frac{1}{2}$ (c) 1 (d) none of these

4. What is the probability that the marble will be any one colour?

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

5. What is the probability that the marble will be red or green?

(a) $\frac{2}{5}$ (b) $\frac{3}{25}$ (c) $\frac{1}{5}$ (d) none of these

6. What is the probability that the marble will be blue?

(a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) 1 (d) 0

Cards are marked with numbers 1 to 50 are placed in the box and mixed thoroughly. One card is drawn at random from the box. Answer the following questions from 7 to 15.

7. What is the probability of getting a number 5?

(a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

8. What is the probability of getting a number less than 11?

(a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

9. What is the probability of getting a number greater than 50?

(a) 1 (b) 0 (c) $\frac{1}{5}$ (d) $\frac{2}{5}$

10. What is the probability of getting a multiple of 5?

(a) 1 (b) 0 (c) $\frac{1}{25}$ (d) $\frac{1}{5}$

11. What is the probability of getting an even number?

(a) 1 (b) $\frac{1}{2}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

12. What is the probability of getting an odd number?

(a) 1 (b) $\frac{1}{2}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

13. What is the probability of getting a prime number?

(a) 1 (b) $\frac{1}{2}$ (c) $\frac{4}{10}$ (d) $\frac{3}{10}$

14. What is the probability of getting a number divisible by 3?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{12}{25}$ (d) $\frac{13}{25}$

15. What is the probability of getting a number divisible by 4?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

16. What is the probability of getting a number divisible by 7?

(a) $\frac{8}{25}$ (b) $\frac{9}{25}$ (c) $\frac{6}{25}$ (d) $\frac{3}{25}$

MCQ WORKSHEET-IV

CLASS X: CHAPTER - 15

PROBABILITY

1. A coin is tossed 1000 times and 560 times a "head" occurs. The empirical probability of occurrence of a Head in this case is

A. 0.5 B. 0.56 C. 0.44 D. 0.056

2. Two coins are tossed 200 times and the following out comes are recorded

HH	HT/TH	TT
56	110	34

What is the empirical probability of occurrence of at least one Head in the above case

A. 0.33 B. 0.34 C. 0.66 D. 0.83

A die is thrown 200 times and the following outcomes are noted, with their frequencies:

Outcome	1	2	3	4	5	6
Frequency	56	22	30	42	32	18

3. What is the empirical probability of getting a 1 in the above case.

A. 0.28 B. 0.22 C. 0.15 D. 0.21

4. What is the empirical probability of getting a number less than 4 ?

A. 0.50 B. 0.54 C. 0.46 D. 0.52

5. What is the empirical probability. of getting a number greater than 4.

A. 0.32 B. 0.25 C. 0.18 D. 0.30

6. On a particular day, the number of vehicles passing a crossing is given below :

Vehicle	Two wheeler	Three wheeler	Four wheeler
Frequency	52	71	77

What is the probability of a two wheeler passing the crossing on that day ?

A. 0.26 B. 0.71 C. 0.385 D. 0.615

7. The following table shows the blood-group of 100 students

Blood group	A	B	O	AB	B^+
Number of Students	12	23	35	20	10

One student is taken at random. What is probability that his blood group is B^+

A. 0.12 B. 0.35 C. 0.20 D. 0.10

8. In a bag, there are 100 bulbs out of which 30 are bad ones. A bulb is taken out of the bag at random. The probability of the selected bulb to be good is

A. 0.50 B. 0.70 C. 0.30 D. None of these

9. On a page of telephone directory having 250 telephone numbers, the Frequency of the unit digits of those number are given below :

0	1	2	3	4	5	6	7	8	9
18	22	32	28	40	30	30	22	18	10

A telephone number is selected from the page at random. What is the probability that its unit digit is

(*a*)2

A. 0.16 B. 0.128 C. 0.064 D. 0.04

(*b*) More than 6

A. 0.20 B. 0.25 C. 0.32 D. 0.16

(*c*) less than 2

A. 0.16 B. 0.18 C. 0.22 D. 0.32

10. **10 defective pens are accidentally mixed with 90 good ones. It is not possible to just look at a pen and tell whether or not it is defective. One pen is taken out at random from this lot. Determine the probability that the pen taken out is a good one.**

A. 0.10 B. 0.20 C. 0.90 D. 1.0

MCQ WORKSHEET-V
CLASS X: CHAPTER - 15
PROBABILITY

One card is drawn from a well-shuffled deck of 52 cards. Answer the question from 1 to 12.

1. Find the probability of getting a king of red colour

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

2. Find the probability of getting a face card.

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{13}$

3. Find the probability of getting a black face card

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

4. Find the probability of getting an ace.

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

5. Find the probability of getting a black card.

(a) $\frac{1}{2}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

6. Find the probability of getting a face card or an ace.

(a) $\frac{4}{13}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{13}$

7. Find the probability of getting face card or black card.

(a) $\frac{4}{13}$ (b) $\frac{8}{13}$ (c) $\frac{7}{13}$ (d) $\frac{3}{13}$

8. Find the probability of getting a king or red card.

(a) $\frac{4}{13}$ (b) $\frac{8}{13}$ (c) $\frac{7}{13}$ (d) $\frac{3}{13}$

9. Find the probability of getting a king and red card.

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

10. Find the probability of getting a king or queen card.

(a) $\frac{1}{26}$ (b) $\frac{2}{13}$ (c) $\frac{1}{13}$ (d) $\frac{3}{26}$

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY : DICE BASED QUESTIONS

1. An unbiased die is thrown. What is the probability of getting

a). an even number
b). a multiple of 3
c). a multiple of 2 or 3
d). a number less than 5 divisible by 2.
e). A number greater than 2 divisible by 3.
f). an even number or a multiple of 3
g). an even number and a multiple of 3
h). a number 3 or 4
i). an odd number
j). a number less than 5
k). a number greater than 3
l). a number between 3 and 6.

2. Two dice are thrown together. Find the probability that the product of the numbers on the top of the dice is (i) 6 (ii) 12 (iii) 7

3. Two dice are thrown at the same time and the product of numbers appearing on them is noted. Find the probability that the product is less than 9.

4. Two dice are numbered 1, 2, 3, 4, 5, 6 and 1, 1, 2, 2, 3, 3, respectively. They are thrown and the sum of the numbers on them is noted. Find the probability of getting each sum from 2 to 9 separately.

5. Two dice are thrown at the same time. Determine the probabiity that the difference of the numbers on the two dice is 2.

6. Two dice are thrown at the same time. Find the probability of getting (i) same number on both dice. (ii) different numbers on both dice.

7. Two dice are thrown simultaneously. What is the probability that the sum of the numbers appearing on the dice is (i) 7? (ii) a prime number? (iii) 1?

8. Two dice are thrown simultaneously. Find the probability of getting

a). an even number on first dice
b). an odd number on first dice
c). an even number as the sum
d). a multiple of 5 as the sum
e). a multiple of 7 as the sum
f). a multiple of 3 as the sum
g). a sum more than 7
h). a sum greater than 9
i). neither the sum 9 nor the sum 11 as the sum
j). a sum less than 6

k). a sum less than 7
l). a sum more than 7
m). a multiple of 3 on one dice
n). a multiple of 2 on one dice
o). a multiple of 5 on one dice
p). a multiple of 2 on one dice and a multiple of 3 on the other
q). a doublet
r). a doublet of even number
s). a doublet of odd number
t). a doublet of prime number
u). a number other than 5 on any dice
v). a number other than 3 on any dice
w). the sum equal to 12.
x). the sum greater than equal to 10
y). the sum less than or equal to 10
z). the sum as a prime number

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY
PLAYING CARDS BASED QUESTIONS

1. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting

(a) an ace card
(b) a red card
(c) either red or king card
(d) red and a king
(e) '2' of spades
(f) '10' of a black suit
(g) a queen of black suit
(h) either a black card or a king
(i) black and king card
(j) a jack, queen or a king
(k) a heart card
(l) a queen card
(m) the ace of spades
(n) the seven of clubs
(o) a ten
(p) a black card
(q) neither a heart nor a king
(r) neither an ace nor a king
(s) neither a red card nor a queen card
(t) a face card or an ace
(u) a face card or a black card
(v) a face card and a black card
(w) neither a face card nor an ace
(x) neither a face card nor '10' card
(y) either a king or red card
(z) either an ace or black card
(aa) an ace and a black card
(bb) a king of red colour card
(cc) a face card
(dd) a red face card
(ee) the jack of hearts
(ff) a spade card
(gg) the queen of diamonds
(hh) '9' of black suit
(ii) a face card or spade card

2. Five cards—the ten, jack, queen, king and ace of diamonds, are well-shuffled with their face downwards. One card is then picked up at random.

(i) What is the probability that the card is the queen?
(ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?

3. The king , queen and jack of clubs are removed from a pack of 52 playing cards. One card is selected at random from the remaining cards. Find the probability that the card is

(a) neither a heart nor a king
(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a king card
(g) a heart card
(h) a red card
(i) a black card
(j) a spade card
(k) a diamond card
(l) a club card
(m) either an ace card or black card
(n) an ace card
(o) a face card
(p) a face card with red colour
(q) neither '10' card nor an ace
(r) an even number card
(s) an odd number card
(t) not a natural number.

6. All spades are removed from a well shuffled deck of 52 cards and then one card is drawn randomly from the remaining cards. Find the probability of getting

(a) neither a heart nor a king
(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a red card
(g) a black card
(h) a spade card
(i) a diamond card
(j) a club card
(k) either an ace card or black card
(l) an ace card
(m) a face card with red colour
(n) neither '10' card nor an ace
(o) an even number card
(p) a face card
(q) an odd number card

4. All face cards are removed from a well shuffled deck of 52 cards and then one card is drawn randomly from the remaining cards. Find the probability of getting

(a) neither a heart nor a king

(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a red card
(g) a black card
(h) a spade card
(i) a diamond card
(j) a club card
(k) either an ace card or black card
(l) an ace card
(m) a face card with red colour
(n) neither '10' card nor an ace
(o) an even number card
(p) an odd number card

5. All cards of ace, jack and queen are removed from a deck of playing cards. One card is drawn at random from the remaining cards, find the probability that the card drawn
(a) neither a heart nor a king
(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a king card
(g) a heart card
(h) a red card
(i) a black card
(j) a spade card
(k) a diamond card
(l) a club card
(m) either an ace card or black card
(n) an ace card
(o) a face card
(p) a face card with red colour
(q) neither '10' card nor an ace
(r) an even number card
(s) an odd number card
(t) not a natural number.

6. All cards of '10', an ace and queen cards are removed from a well shuffled deck of 52 cards and then one card is drawn randomly from the remaining cards. Find the probability of getting
(a) neither a heart nor a king
(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a king card
(g) a heart card

(h) a red card
(i) a black card
(j) a spade card
(k) a diamond card
(l) a club card
(m) either an ace card or black card
(n) an ace card
(o) a face card
(p) a face card with red colour
(q) neither '10' card nor an ace
(r) an even number card
(s) an odd number card
(t) not a natural number.

7. Five cards—the ten, jack, queen, king and ace of diamonds, are removed from the well-shuffled 52 playing cards. One card is then picked up at random. Find the probability of getting

(a) neither a heart nor a king
(b) neither an ace nor a king
(c) neither a red card nor a queen card
(d) a black card or an ace.
(e) either a heart or a spade card
(f) a king card
(g) a heart card
(h) a red card
(i) a black card
(j) a spade card
(k) a diamond card
(l) a club card
(m) either an ace card or black card
(n) an ace card
(o) a face card
(p) a face card with red colour
(q) neither '10' card nor an ace
(r) an even number card
(s) an odd number card
(t) not a natural number.

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY : COINS BASED QUESTIONS

1. Two coins are tossed simultaneously. Find the probability of getting
- **i).** at least one head
- **ii).** at most one head
- **iii).** exactly two head
- **iv).** exactly one head
- **v).** no head
- **vi).** no tail
- **vii).** at least one tail
- **viii).** at most one tail
- **ix).** exactly two tails
- **x).** exactly one tail

2. A coin is tossed two times. Find the probability of getting at most one head.

3. A coin is tossed 3 times. List the possible outcomes. Find the probability of getting (i) all heads (ii) at least 2 heads

4. Sushma tosses a coin 3 times and gets tail each time. Do you think that the outcome of next toss will be a tail? Give reasons.

5. If I toss a coin 3 times and get head each time, should I expect a tail to have a higher chance in the 4th toss? Give reason in support of your answer.

6. Three coins are tossed simultaneously. What is the probability of getting
- **i).** exactly two heads
- **ii).** at least two heads
- **iii).** at most two heads
- **iv).** one head or two heads
- **v).** exactly one tail
- **vi).** at least one tail
- **vii).** at most one tail
- **viii).** at least two tails
- **ix).** at most two tails
- **x).** exactly two tails
- **xi).** no head
- **xii).** no tail

7. Four coins are tossed simultaneously. What is the probability of getting
- **i).** exactly one head
- **ii).** exactly two heads
- **iii).** exactly three heads
- **iv).** at least one head
- **v).** at most one head

vi). at least three heads
vii). at most three heads
viii). at least two heads
ix). at most two heads
x). one head or two heads
xi). exactly one tail
xii). at least one tail
xiii). at most one tail
xiv). at least two tails
xv). at most two tails
xvi). at least three tails
xvii). at most three tails
xviii). exactly two tails
xix). no head
xx). no tail

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY
BAG BALLS BASED QUESTIONS

1. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is

1. red ?
2. not red?

2. A bag contains 5 red, 8 green and 7 white balls. One ball is drawn at random from the bag, find the probability of getting

3. a white ball or a green ball
4. neither green ball nor red ball.
5. not green?
6. not red?
7. not white?
8. neither red ball nor white ball?

3. A bag contains 5 red balls, 8 white balls and 4 green balls. One ball is taken out of the bag at random. What is the probability that the ball taken out will be

9. red ?
10. white ?
11. not green?
12. not red?
13. not white?
14. neither red ball nor white ball?

4. A box contains 3 blue, 2 white, and 4 red balls. If a ball is drawn at *random* from the box, what is the probability that it will be

15. white?
16. blue?
17. red?
18. neither blue ball nor red ball?
19. neither blue ball nor white ball?
20. neither white ball nor red ball?
21. not blue?
22. not red?
23. not white?

5. A bag contains 4 blue, 5 black, 6 red and 5 white balls. One ball is taken out of the bag at random. What is the Probability that it will be

24. black?
25. blue?
26. red?
27. white?
28. black or blue?
29. white or blue?

30. red or blue?
31. white or red?
32. neither blue ball nor red ball?
33. neither red ball nor white ball?
34. neither blue ball nor black ball?
35. not blue ball?
36. not red ball?
37. not white ball?
38. not black ball?

6. A bag contains 9 blue, 4 black, 5 red and 7 white balls. One ball is taken out of the bag and found red ball then again one ball is taken out at random from the remaining. What is the Probability that it will be

39. black?
40. blue?
41. red?
42. white?
43. black or blue?
44. white or blue?
45. red or blue?
46. white or red?
47. neither blue ball nor red ball?
48. neither red ball nor white ball?
49. neither blue ball nor black ball?
50. not blue ball?
51. not red ball?
52. not white ball?
53. not black ball?

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY : NUMBER BASED QUESTIONS

1. On one page of a telephone directory, there were 200 telephone numbers. The frequency distribution of their unit place digit (for example, in the number 25828573, the unit place digit is 3) is given in below table:

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	22	26	22	22	20	10	14	28	16	20

Without looking at the page, the pencil is placed on one of these numbers, i.e., the number is chosen at *random*. What is the probability that the digit in its unit place is (i) an odd number (ii) a prime number and (iii) a number greater than 4.?

2. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). neither divisible by 5 nor 10.
n). neither divisible by 2 nor 5.
o). neither divisible by 3 nor 5.
p). a perfect cube number.
q). a prime number
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

3. Cards are marked with numbers 4, 5, 6, …….50 are placed in the box and mixed thoroughly. One card is drawn at random from the box. What is the probability of getting

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.

l). a two digit number perfect square.
m). neither divisible by 5 nor 10.
n). neither divisible by 2 nor 5.
o). neither divisible by 3 nor 5.
p). a perfect cube number.
q). a prime number
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

4. Cards are marked with numbers 13, 14, 15, …….60 are placed in the box and mixed thoroughly. One card is drawn at random from the box. What is the probability of getting

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

5. There are 30 cards numbered from1 to 30. One card is drawn at random. Find the probability of getting the card with

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.

p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

6. A box contains 25 cards numbered from 1 to 25. A card is drawn from the box at random. Find the probability of getting the card with

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

7. A box contains 19 balls bearing numbers 1,2,3, …. 19 respectively. A ball is drawn at random from the box, Find the probability that the number on the ball is

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.

t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

8. A box contains 20 balls bearing numbers 1,2,3, …. 20 respectively. A ball is drawn at random from the box, Find the probability that the number on the ball is

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

9. 15 cards numbered 1, 2, 3, 4,……. 14, 15 are put in a box and mixed thoroughly. A man draws a card at random from the box. Find the probability that the number on the card is

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

10. Tickets numbered 2, 3, 4, 5, …..100, 101 are placed in a box and mixed thoroughly. One ticket is drawn at random from the box. Find the probability that the number on the ticket is

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

11. Cards are marked with numbers 5, 6, 7, …….50 are placed in the box and mixed thoroughly. One card is drawn at random from the box. What is the probability of getting

a). a two-digit number
b). a perfect square number
c). a number divisible by 5.
d). a number divisible by 2 or 3.
e). a number divisible by 2 and 3.
f). a number divisible by 7.
g). a number multiple of 8.
h). a two digit number divisible by 5.
i). a two digit number divisible by 2.
j). a two digit number divisible by 3.
k). a two digit number divisible by 4.
l). a two digit number perfect square.
m). a perfect cube number.
n). a prime number.
o). neither divisible by 5 nor 10.
p). neither divisible by 2 nor 5.
q). neither divisible by 3 nor 5.
r). a two digit prime number.
s). an even prime number.
t). a number is not divisible by 5.
u). a number is not divisible by 3.
v). a number is not divisible by 2 and 3.

PRACTICE QUESTIONS
CLASS X : CHAPTER – 15
PROBABILITY
LOGICAL REASONING BASED QUESTIONS

1. Why is tossing a coin considered to be a fair way of deciding which team should get the ball at the beginning of a football game?

2. The probability that it will rain today is 0.84. What is the probability that it will not rain today?

3. What is the probability that an ordinary year has 53 Sundays?

4. Find the probability of getting 53 Fridays in a leap year.

5. Find the probability of getting 53 Fridays or 53 Saturdays in a leap year.

6. Find the probability of getting 53 Mondays or 53 Tuesday in an ordinary year.

7. Out of 400 bulbs in a box, 15 bulbs are defective. One bulb is taken out at random from the box. Find the probability that the drawn bulb is not defective.

8. In a lottery there are 10 prizes and 25 blanks. What is the probability of getting a prize?

9. 250 lottery tickets were sold and there are 5 prizes on these tickets. If Mahesh purchased one lottery ticket, what is the probability that he wins a prize?

10. The record of a weather station shows that out of the past 250 consecutive days, its weather forecasts were correct 175 times. (i) What is the probability that on a given day it was correct? (ii) What is the probability that it was not correct on a given day?

11. A lot consists of 144 ball pens of which 20 are defective and the others are good. Nuri will buy a pen if it is good, but will not buy if it is defective. The shopkeeper draws one pen at random and gives it to her. What is the probability that (i) She will buy it ? (ii) She will not buy it ?

12. A box contains 5 red marbles, 8 white marbles and 4 green marbles. One marble is taken out of the box at random. What is the probability that the marble taken out will be (i) red ? (ii) white ? (iii) not green?

13. Savita and Hamida are friends. What is the probability that both will have (i) different birthdays? (ii) the same birthday? (ignoring a leap year).

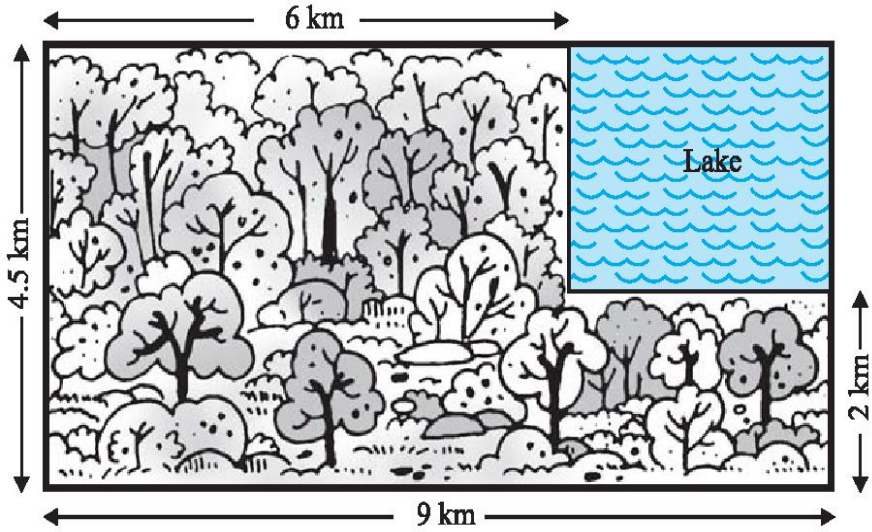
14. 12 defective pens are accidentally mixed with 132 good ones. It is not possible to just look at a pen and tell whether or not it is defective. One pen is taken out at random from this lot. Determine the probability that the pen taken out is a good one.

15. A piggy bank contains hundred 50p coins, fifty Re 1 coins, twenty Rs 2 coins and ten Rs 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a 50 p coin ? (ii) will not be a Rs 5 coin?

16. Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish. What is the probability that the fish taken out is a male fish?

17. A number x is selected from the numbers 1, 2, 3 and then a second number y is randomly selected from the number 1, 4, 9. What is the probability that the product xy of the two numbers will be less than 9?

18. A missing helicopter is reported to have crashed somewhere in the rectangular region shown in Fig. What is the probability that it crashed inside the lake shown in the figure?



19. There are 40 students in Class X of a school of whom 25 are girls and 15 are boys. The class teacher has to select one student as a class representative. She writes the name of each student on a separate card, the cards being identical. Then she puts cards in a bag and stirs them thoroughly. She then draws one card from the bag. What is the probability that the name written on the card is the name of (i) a girl? (ii) a boy?

20. A carton consists of 100 shirts of which 88 are good, 8 have minor defects and 4 have major defects. Jimmy, a trader, will only accept the shirts which are good, but Sujatha, another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. What is the probability that (i) it is acceptable to Jimmy? (ii) it is acceptable to Sujatha?

21. Two customers Shyam and Ekta are visiting a particular shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on (i) the same day? (ii) consecutive days? (iii) different days?

22. Two customers are visiting a particular shop in the same week (Monday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on (i) the same day? (ii) consecutive days? (iii) different days?

23. A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is *double* that of a red ball, determine the number of blue balls in the bag.

24. A box contains 12 balls out of which *x* are black. If one ball is drawn at random from the box, what is the probability that it will be a black ball? If 6 more black balls are put in the box, the probability of drawing a black ball is now double of what it was before. Find *x*.

25. A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$. Find the number of blue marbles in the jar.

26. A lot consists of 144 ball pens of which 20 are defective and the others are good. Nuri will buy a pen if it is good, but will not buy if it is defective. The shopkeeper draws one pen at random and gives it to her. What is the probability that (i) She will buy it ? (ii) She will not buy it ?

27. A jar contains 54 marbles, each of which is blue, green or white. If a marble is drawn at random from the jar, the probability that it is green is $\frac{1}{3}$ and that of getting a blue marble is $\frac{4}{9}$. Find the number of white marbles in the jar.

28. A letter is chosen at random from the letters of the word 'ASSASSINATION'. Find the probability that the letter chosen is a (i) vowel (ii) consonant (iii) A (iv) S (v) N.

29. A letter is chosen at random from the letters of the word 'INDEPENDENCE'. Find the probability that the letter chosen is a (i) vowel (ii) consonant (iii) E (iv) N (v) D.

30. A letter is chosen at random from the letters of the word 'MATHEMATICS'. Find the probability that the letter chosen is a (i) vowel (ii) consonant (iii) A (iv) T (v) M.

31. A letter of English alphabets is chosen at random. Determine the probability that the letter is a consonant.

32. There are 1000 sealed envelopes in a box, 10 of them contain a cash prize of Rs 100 each, 100 of them contain a cash prize of Rs 50 each and 200 of them contain a cash prize of Rs 10 each and rest do not contain any cash prize. If they are well shuffled and an envelope is picked up out, what is the probability that it contains no cash prize?

33. Box A contains 25 slips of which 19 are marked Re 1 and other are marked Rs 5 each. Box B contains 50 slips of which 45 are marked Re 1 each and others are marked Rs 13 each. Slips of both boxes are poured into a third box and reshuffled. A slip is drawn at random. What is the probability that it is marked other than Re 1?

34. A carton of 24 bulbs contain 6 defective bulbs. One bulbs is drawn at random. What is the probability that the bulb is not defective? If the bulb selected is defective and it is not replaced and a second bulb is selected at random from the rest, what is the probability that the second bulb is defective?

35. A child's game has 8 triangles of which 3 are blue and rest are red, and 10 squares of which 6 are blue and rest are red. One piece is lost at random. Find the probability that it is a (i) triangle (ii) square (iii) square of blue colour (iv) triangle of red colour

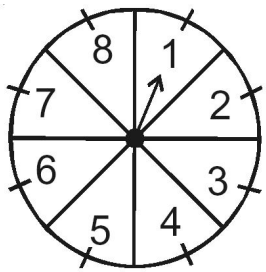
36. In a game, the entry fee is Rs 5. The game consists of a tossing a coin 3 times. If one or two heads show, Sweta gets her entry fee back. If she throws 3 heads, she receives double the entry fees. Otherwise she will lose. For tossing a coin three times, find the probability that she (i) loses the entry fee. (ii) gets double entry fee. (iii) just gets her entry fee.

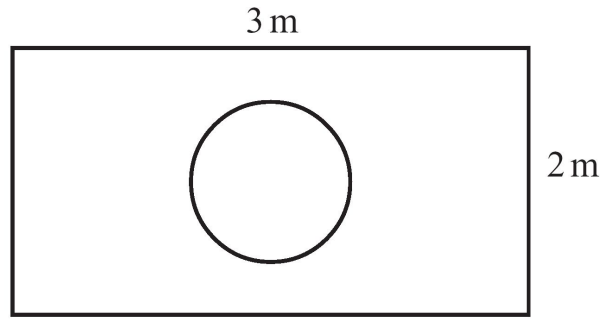
37. A die has its six faces marked 0, 1, 1, 1, 6, 6. Two such dice are thrown together and the total score is recorded. (i) How many different scores are possible? (ii) What is the probability of getting a total of 7?

38. A lot consists of 48 mobile phones of which 42 are good, 3 have only minor defects and 3 have major defects. Varnika will buy a phone if it is good but the trader will only buy a mobile if it has no major defect. One phone is selected at random from the lot. What is the probability that it is (i) a good phone (ii) a bad phone

39. (i) A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?
(ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective ?

40. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 (see Fig.), and these are equally likely outcomes. What is the probability that it will point at (i) 8 ? (ii) an odd number? (iii) a number greater than 2? (iv) a number less than 9?





41. Suppose you drop a die at random on the rectangular region shown in above right-sided figure. What is the probability that it will land inside the circle with diameter 1m?

42. A child has a die whose six faces show the letters as given below:

A	B	C	D	E	A

The die is thrown once. What is the probability of getting (i) A? (ii) D?

43. A game consists of tossing a one rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game.